

Filtered Workbook Analysis & Insights

Interactive Data Slicing & Key Business Questions Resolution

1,000

TOTAL SAMPLE SIZE

481

BIKE BUYERS

3 Regions

GEOGRAPHIC SLICES

5 Questions

EXECUTIVE ANSWERS

1. Filtered Workbook Architecture

The 3. Dashboard.xlsx workbook contains dynamic slicers that allow real-time filtering across three main dimensions: **Region**, **Marital Status**, and **Education Level**. This page details how filtering impacts subset calculations.

1.1 Filter State Comparison Table

Filter Scenario / Slice	Sample Size (N)	Bike Buyers	Conversion Rate	Avg Income (Buyers)
Unfiltered Dataset (Baseline)	1,000	481	48.1%	\$58,100
Filter: Region = North America	508	220	43.3%	\$58,400
Filter: Region = Europe	300	148	49.3%	\$53,800
Filter: Region = Pacific	192	113	58.9%	\$64,200
Filter: Marital Status = Single	461	251	54.4%	\$54,300
Filter: Education = Bachelors	306	169	55.2%	\$62,800

Filtered Insight: Highest Converting Region

When applying the **Pacific** region slicer, conversion jumps to **58.9%** (vs. 43.3% in North America), driven by higher average buyer incomes (\$64,200) and compact city layouts favoring bike commutes.

2. Executive Questions & Data Answers (Q1 & Q2)

Question 1: How does income level influence bike purchases across different genders?

Business Context: Determines whether marketing messaging should adjust price point positioning based on male vs. female income thresholds.

Data Analysis & Income Breakdown

Analyzing average income across buyers versus non-buyers reveals a consistent income premium required for bike purchasing across both male and female demographics.

Gender	Purchase Status	Average Income (\$)	Income Variance (\$)	Impact on Conversion
Female	No (Non-Buyer)	\$53,440	+\$2,334 (+4.4%)	Moderate correlation between female income and purchasing.
Female	Yes (Buyer)	\$55,774		
Male	No (Non-Buyer)	\$56,208	+\$3,916 (+7.0%)	Stronger income elasticity observed among male buyers.
Male	Yes (Buyer)	\$60,124		

Answer Summary: Higher average income positively correlates with bike purchases across both genders. Male buyers have a higher income baseline (\$60,124) than female buyers (\$55,774), but both buyer segments consistently out-earn their non-purchasing counterparts.

Question 2: What commute distance bracket yields the highest bike purchase volume?

Business Context: Identifies urban vs. suburban targeting locations based on customer daily travel patterns.

Commute Proximity Table

Commute Distance	Non-Buyers	Buyers	Total	Conversion %
0–1 Miles	166	200	366	54.6%
1–2 Miles	92	77	169	45.6%
2–5 Miles	67	95	162	58.6%
5–10 Miles	116	76	192	39.6%
10+ Miles	78	33	111	29.7%

Answer Summary: Customers commuting **0–1 miles** represent the largest absolute volume of buyers (200 buyers), while 2–5 mile commuters achieve the highest relative conversion rate (58.6%). Demand drops steeply for commutes over 5 miles.

3. Executive Questions & Data Answers (Q3 & Q4)

Question 3: Which age bracket dominates total bike sales, and why?

Business Context: Guides demographic targeting for advertising creative, imagery, and channel placement.

Age Group Purchasing Distribution

Age Group	Age Bounds	Buyers (Yes)	Non-Buyers (No)	% of Total Sales
Adolescent	< 31 Years	39	71	8.1%
Middle Age	31–54 Years	383	318	79.6%
Old	55+ Years	59	130	12.3%

Answer Summary: The **Middle Age (31–54)** bracket overwhelmingly dominates bike purchases, accounting for **79.6% (383 out of 481)** of all sales. This segment combines career income stability with active lifestyle preferences, outperforming younger and older age brackets combined.

Question 4: Does marital status impact bike purchase behavior?

Business Context: Informs household-level marketing strategies and bundled family promotional offers.

Marital Status Breakdown

Marital Status	Non-Buyers	Buyers	Total	Conversion Rate
Single	210	251	461	54.4%
Married	309	230	539	42.7%

Answer Summary: **Single customers** convert at a significantly higher rate (**54.4%**) compared to married customers (**42.7%**). Single individuals likely face fewer household financial trade-offs and show higher inclination toward personal commuting equipment.

4. Executive Question 5 & Filtered Action Strategy

Question 5: How do education level and region interact when filtered on the workbook dashboard?

Business Context: Identifies regional sub-segments to optimize direct mail and localized digital ad targeting.

Filtered Pivot Cross-Tabulation (Education vs. Region Buyers)

Education Level	North America	Europe	Pacific	Total Buyer Volume
Bachelors	72	51	46	169
Partial College	55	38	26	119
High School	48	32	18	98
Graduate Degree	32	20	18	70
Partial High School	13	7	5	25

Answer Summary: Applying combined regional and education filters demonstrates that customers with a **Bachelors degree in North America and Europe** form the single largest buyer cluster (123 buyers), prioritizing premium commuter bike options.

5. Summary Action Plan Based on Filtered Analysis

- **Primary Audience Target:** Middle-aged single professionals (31–54) with annual income over \$55k.
- **Geographic Focus:** Short-distance urban corridors (0–2 miles commute) in Pacific and North American metros.
- **Campaign Slicing Strategy:** Utilize dynamic filtered sub-lists in CRM campaigns to deliver tailored offers based on education level and regional buyer preferences.